Mälardalen University
School of Innovation Design and Engineering
Västerås, Sweden

Master Thesis Proposal
Course Code: XXX000

PRELIMINARY TITLE FOR THE THESIS

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Company name, location

17th June 2026

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[1] ***[MUST HAVE BEEN DISCUSSED WITH THEM BEFORE SUBMITTED THE PROPOSAL TO THE EVALUATION COMMITTEE. SUPERVISORS AND EXAMINERS CANNOT BE ADDED WITHOUT THEIR CONSENT]***

Problem formulation:

[PROVIDE A BRIEF DESCRIPTION OF WHAT PROBLEM YOU ARE GOING TO INVESTIGATE IN YOUR THESIS WORK. PROVIDE 2 TO 4 CLEAR AND CONCRETE PROBLEMS THAT WILL BE THE CORE OF THE THESIS WORK.]

Method:

[DESCRIBE IN DETAIL THE STEPS YOU WILL TAKE IN ATTEMPTING TO ANSWER YOUR RESEARCH QUESTION OR WHETHER YOU WILL FOLLOW A KNOWN SCIENTIFIC OR ENGINEERING METHOD]

The following requirements are copied verbatim:

- Kretskortsfräsar (stöd av Majid)
  Samtliga kretskortsfräsar ska vara genomgångna, servade och funktionstestade. En komplett och lättillgänglig manual ska vara framtagen och tillgänglig för studenter.
- Issues (LIMB ? GitHub) (Stöd av Johan)
  Issues-listan för LIMB-projektet på GitHub ska vara genomgången, utvärderad och prioriterad utifrån projektets mål och tidsram.
- Tillverkning av HR-EMG-enheter (Stöd av Mahlin)
  Minst 8 stycken högupplösta EMG-enheter med integrerad IMU (HR-EMG) ska vara tillverkade och kapslade. Designen baseras på underlag från Malin Jansson.
- Kommunikation och integration
  CAN-integration för IMU ska vara implementerad i LIMB-systemet. Bluetooth-kommunikation mellan LIMB och HR-EMG-enheter ska vara implementerad och verifierad.
- Mekanisk utvärdering
  En utvärdering ska genomföras avseende ersättning av komponenter i armen (t.ex. kolfiberlösningar) för att förbättra stabilitet och tillförlitlighet.
- Visionssystem
  Kamera ska integreras för att möjliggöra identifiering av kaffekopp samt mänsklig arm, inklusive grundläggande funktionstest.
- ROS2
  En utvärdering ska genomföras av möjligheten att implementera ROS2 i systemet, inklusive för- och nackdelar.
- Projektorganisation
  Gruppkontrakt ska vara etablerat och en tydlig ansvarsfördelning inom gruppen ska vara definierad och dokumenterad.

Outcomes:

Division

The project is divided in optional and mandatory deliverables.

Mandatory deliverables

- Create validate and capsuled 8 IMU PCBS
- Create a very specific step by step manual for the wegstr

Optional deliverables

- Create inverse kinematics for the limb arm movement
- Create new parts for the arm made of carbon fiber to solve issues with movement.
- re design xxx to make sure parts inside fit correctly
- create and by step guide on how to use the 3d printer bambulabs and the laser cutter.
- Solve a proper stand that is able to integrate the power supply and cover remaining cables sticking out.

note: manulas have to be on a extremely basic level where if it is followed anyone should be able to use the machines without a single issue.
note: One of the components does not fit without cracking the arm this is the part that needs to be re designed.
note:the Stand has to be made with consideriation of the arm movement at the moment the stand is in the way and the arm cannot move all the way to 0 degrees and is right now sticking out at a small angle

[EXPAND ON THE PROBLEM FORMULATION BY DESCRIBING WHAT YOU HOPE TO ACCOMPLISH DURING THE THESIS WORK, AND THE DESIRED OUTCOMES (ESPECIALLY THE PRACTICAL OR THEORETICAL BENEFITS TO BE GAINED FROM THE WORK)] In the end we want 8 complete validated and capsuled IMU devices. Each and every member should understand how the modules work and be able to test them with the LIMB robot and be able to control the limb robot without the modules.

Initial time plan:

[DESCRIBE THE ACTIVITIES THAT MUST BE PERFORMED DURING THE WORK, IN WHICH ORDER AND HOW LONG TO YOU PLAN TO HAVE THEM. IT IS NOT ENOUGH TO COPY-PASTE THE IMPORTANT DATES FOR THE COURSE.]

Limitations:

[DESCRIBE CONDITIONS BEYOND YOUR CONTROL THAT PLACE RESTRICTIONS ON WHAT YOU CAN DO AND THE CONCLUSIONS YOU MAY BE ABLE TO DRAW. ALSO, IF THE PROBLEM IS REALLY BROAD, DESCRIBE THE BOUNDARIES THAT YOU WILL FOCUS ON]

For thesis with confidential data, how will this be addressed:

[DESCRIBE HOW THIS ISSUE WILL BE DEALT WITH. IF THE THESIS WORK IS DONE WITH A COMPANY, THIS ISSUE MUST HAVE BEEN DISCUSSED AND THE PROPOSED SOLUTION MUST HAVE BEEN AGREED BY THE COMPANY BEFORE SUBMITTING THE PROPOSAL.]

[REMOVE ALL THE INSTRUCTIONS MARKED WITH RED COLOR BEFORE SUBMITTING YOUR PROPOSAL.]